

SANTA YNEZ, CA, USA

WMO: 723762

Lat: 34.607N

Lon: 120.076W

Elev: 205

StdP: 98.89

Time Zone: -8.00 (NAP)

Period: 09-19

WBAN: 93244

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.3	0.2	-8.7	1.8	16.0	-7.0	2.1	15.6	9.1	13.6	8.1	13.5	0.7	70	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	35.9	18.9	32.9	18.1	32.0	18.1	20.4	31.6	19.5	30.8	18.8	30.0	4.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.8	24.6	15.1	11.0	23.1	14.0	10.2	21.6	59.0	31.5	56.2	30.5	53.7	29.6	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.6	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-4.3	21.7	1.5	1.1	-5.3	22.5	-6.2	23.2	-7.0	23.8	-8.1	24.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	11.2	11.2	12.9	14.8	16.1	18.6	20.8	21.0	20.6	17.7	13.4	10.7
	DBStd	4.64	2.73	3.17	2.86	2.55	2.32	2.52	2.17	2.10	2.91	2.98	3.10	2.93
	HDD10.0	87	19	23	7	1	0	0	0	0	0	0	8	29
	HDD18.3	1244	221	199	169	108	77	24	3	1	8	48	149	238
	CDD10.0	2202	58	57	97	147	189	259	334	342	318	239	111	49
	CDD18.3	315	0	0	2	3	8	33	78	85	76	28	3	0
	CDH23.3	5257	27	47	99	197	280	644	1066	1095	1039	587	164	12
	CDH26.7	2224	2	4	25	52	104	262	465	503	504	253	49	1
Wind	WSAvg	2.2	1.8	2.1	2.3	2.6	2.8	2.6	2.4	2.2	2.0	1.8	1.7	1.9
Precipitation	PrecAvg	438	94	97	71	31	11	2	1	0	2	21	34	73
	PrecMax	852	416	405	324	115	65	17	7	2	12	106	112	238
	PrecMin	172	6	4	1	3	1	0	0	0	0	1	4	7
	PrecStd	169	91	85	72	29	14	5	2	1	3	24	28	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	27.3	30.0	31.2	32.8	36.4	36.2	37.3	37.9	36.3	31.3	25.1
		MCWB	11.4	13.5	14.9	15.2	15.7	18.8	19.6	19.4	18.7	16.6	14.7	12.3
	2%	DB	22.9	25.0	26.2	27.8	29.9	32.6	33.0	34.8	35.0	32.5	27.6	22.1
		MCWB	11.6	13.1	14.1	14.7	16.1	18.5	19.1	19.3	18.7	16.3	13.4	11.8
	5%	DB	21.2	22.1	22.8	25.9	26.3	29.7	32.0	32.1	32.4	29.9	24.9	19.0
		MCWB	11.6	12.4	13.5	14.1	15.3	17.7	19.1	19.2	18.2	15.4	12.4	12.0
	10%	DB	17.9	17.9	20.8	22.7	22.9	27.2	29.5	29.8	29.7	26.9	22.1	17.1
		MCWB	11.0	11.7	13.1	13.7	14.6	16.8	18.4	18.5	17.7	14.6	11.9	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	15.8	17.7	18.2	18.0	20.2	20.9	21.5	21.7	19.6	15.8	15.6
		MCDB	21.1	20.2	24.9	26.6	29.4	32.9	32.6	33.4	33.1	27.8	25.3	20.5
	2%	WB	13.6	14.7	15.7	16.3	16.7	19.0	20.1	20.1	20.1	18.0	14.8	14.4
		MCDB	18.7	20.2	22.4	23.9	27.0	30.4	31.7	31.6	30.6	27.9	23.1	18.1
	5%	WB	12.6	13.6	14.6	15.3	15.7	18.0	19.2	19.1	18.9	16.9	13.9	13.1
		MCDB	17.8	20.0	21.0	23.1	24.9	28.7	30.5	30.9	30.0	25.9	21.3	16.5
	10%	WB	11.8	12.4	13.5	14.3	14.8	17.1	18.3	18.3	18.0	15.8	12.9	12.0
		MCDB	16.0	17.8	19.3	21.3	23.2	26.8	28.8	29.5	28.3	24.1	19.6	15.2
Mean Daily Temperature Range	5% DB	MDBR	14.7	14.3	14.4	15.9	14.9	16.7	17.6	17.9	18.0	17.7	16.8	13.6
		MCDBR	19.9	20.5	19.8	21.1	20.5	22.0	21.1	21.8	21.9	23.2	22.8	17.3
		MCWBR	10.5	10.7	10.0	9.6	9.0	8.8	8.3	8.2	8.3	9.4	10.5	9.2
	5% WB	MCDBR	15.0	16.4	16.2	17.0	17.6	19.4	19.7	19.8	18.8	17.9	18.3	12.1
		MCWBR	9.0	9.3	9.2	8.9	8.3	8.5	8.0	8.0	7.7	7.8	9.3	7.3
Clear-Sky Solar Irradiance	taub	0.295	0.309	0.315	0.346	0.360	0.348	0.372	0.353	0.349	0.337	0.298	0.300	
	taud	2.603	2.577	2.563	2.451	2.433	2.470	2.451	2.473	2.492	2.524	2.613	2.601	
	Ebn at Noon	924	945	962	939	924	930	905	921	912	898	914	896	
	Edh at Noon	76	86	95	112	115	111	112	108	101	90	74	72	
All-Sky Solar Radiation	RadAvg	3.03	3.90	5.30	6.62	7.35	7.65	7.34	6.83	5.85	4.49	3.42	2.73	
	RadStd	0.24	0.41	0.34	0.38	0.46	0.34	0.27	0.26	0.25	0.32	0.18	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.60	N/A	-1.45	+0.67	N/A	N/A	N/A	-128	+218	+82

Nomenclature: See separate page